FORWARD DEPLOYED AI ENGINEERING

Capstone Project

Stage One: Discovery

The workbook you complete before you design anything

Fill this in first. Nothing in the later stages is valid unless it can be traced back to something you recorded on these pages.

WEEK ONE · DAYS ONE TO THREE

How to use this workbook

This workbook is a set of tables for you to complete. The prompts in the left-hand column tell you what belongs in each row; the remaining columns are yours to fill in. Where a table has blank rows at the bottom, add as many more as you need.

Work through the sections in order. Section one establishes who you are talking to, sections two to five gather the evidence, and section six is where you turn that evidence into a problem statement. Do not attempt section six before the others are genuinely finished, because the whole point of the exercise is that the conclusion follows from the evidence rather than the other way round.

What finished looks like

You are finished with discovery when a further interview tells you nothing you did not already know, and when every claim in your problem statement can be pointed at a specific row in one of these tables. If either of those is not yet true, you are not finished, however much time has passed.

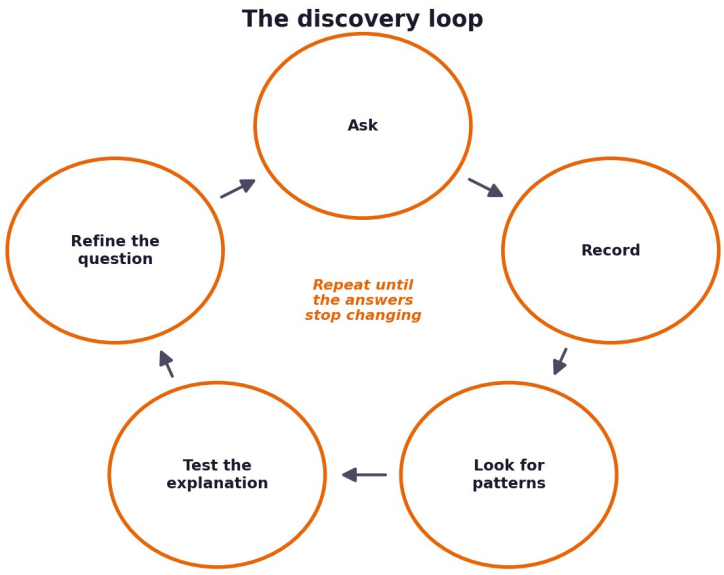


Figure 1. Discovery repeats until the answers stop changing.

1 Section one: what the interviews tell you

Your discovery evidence is in two places: the five stakeholder interview transcripts in the datasets folder, and the ticket data itself. Start with the transcripts. Read all five before you write anything, then come back and fill in the table below.

The prompts in the left column are the people you spoke to. For each, record what you were trying to learn from them, what they actually told you, and, importantly, what they seemed not to know about their own operation. That last column is often where the useful material is, because each of the five can only see their own part of the problem.

Who	What they told you	What they appear not to know	What you want to verify in the data
Marcus Adeyemi, Head of Support	>500 tickets/week; six agents; 8–12h replies vs 2h SLA; FCR around 42%; escalation costs ~4x. Wants safe automation.	Distribution and answerability estimates are not measured outcomes.	Historical FCR, doc answerability, route mix; live first reply time unavailable.
Sofia Restrepo, Tier One Agent	40–70 tickets/day; familiar cases 4–5 min; unusual cases up to 40 min; searches are poor; escalates uncertain/security cases.	Her estimate that ~70% are familiar and non-fluent customers fare worse.	Doc-answerable share and fluency-group historical outcomes.
Daniel Okafor, Tier Two Engineer	About half of escalations might be Tier 1 solvable with the right page; wants summary, sources, and uncertainty.	Estimate is experience, not labeled data.	Retrieval expected-source hit and safe response coverage.
Ines Carvalho, Technical Writer	29 regularly reviewed articles; exact-title search is poor; personal snippets are unreviewed; wants source IDs.	Documentation may still fail to cover roadmap/novel incidents.	Compare article corpus with ticket expected sources; no quality audit supplied.
Ravi Shah, Customer	Slow replies; self-search finds answer about half the time; wants honest drafts with sources, urgency-sensitive handling.	Perceived self-search and tier reply experience lack direct telemetry.	Historical answerability, CSAT and FCR; no reply timestamp by tier.

Where the accounts disagree

At least three points of disagreement run through the transcripts, and each can be settled by going to the ticket data rather than by deciding whose account sounds more convincing. Identify them, then resolve them with evidence.

The disagreement	Who says what	How the data settles it	What follows from the answer
Can docs answer most tickets?	Ines: corpus is sound; Sofia: search fails; Marcus estimates automation potential.	357/500 development tickets labeled answerable (71.4%); this indicates corpus match labels, not successful self-service.	Build retrieval over reviewed articles, cite sources, escalate unsupported cases.
Do non-fluent customers have worse outcomes?	Sofia suspects lower CSAT; customer worries about access.	Dev data: non-fluent FCR 45.8% vs fluent 43.2%; CSAT 3.04 vs 2.95; no worse historical outcome signal.	Do not assert her hypothesis; monitor live quality by language.
Are escalations avoidable?	Daniel estimates about half; Marcus notes high cost.	Labels expect auto route for 311/500 (62.2%); only 219/500 historical FCR (43.8%). Not directly comparable to escalation solvability.	Measure safe containment and human rework in pilot.
Does enterprise wait longer?	Ravi reports urgency/tier effects; no timestamp evidence in pack.	No actual first-response timestamp by tier is present.	Keep as open telemetry requirement; do not claim measured SLA improvement.

What nobody said

Real stakeholders leave out the things so familiar to them that they no longer notice them. Record anything you expected somebody to raise that none of the five actually did.

What was never mentioned	Why you would have expected it	How you will check it
Live first reply time by tier/channel	Client SLA and customer experience depend on timeliness.	Add event timestamps in pilot; pack only has historical resolution

		time.
Human rating of generated answer quality	Governance requires hallucination and citation assessment.	Two independent assessors rate at least 50; not completed in this run.
Customer consent, retention and deletion	Ticket text can contain personal information.	Confirm data policy with client; use minimization, access control and retention schedule.
Workload/capacity actuals	The sprint plan depends on personal availability.	Kaushik to provide actual hours/dates; do not infer.

2 Section two: what the ticket data shows

Before you interpret anything, count it. Load development_tickets.json and fill in the table below from the data itself rather than from impression. Where a figure surprises you, write the surprise down; those are usually the most productive threads to pull.

What to measure	Your figure	Where the figure came from	What surprised you about it
Total tickets in development sample	500	development_tickets.jso n	Large enough for broad baselines, still synthetic/project-specific.
Split by channel	email 212 (42.4%); chat 155 (31.0%); docs_comment 78 (15.6%); forum 55 (11.0%)	Development dataset	Email is largest, rather than chat.
First-contact resolution	219/500 = 43.8%	history.first_contact_res olved	Below the brief's 60% goal; differs from brief's approximate 42%.
Mean historical CSAT	2.972/5	history.csat	Lower than Project Brief's 3.2 narrative baseline.
Historical repeat contacts	108/500 = 21.6%	history.repeat_contact	Useful baseline, but synthetic history.
Documentation answerable	357/500 = 71.4%	labels.answerable_from_ docs	Opportunity exists; not proof a response is safe or usable.
Expected routing	311 auto / 189 escalation	labels.expected_route	Expected labels are not measured model performance.
Urgency mix	high 146; medium 226; low 128	labels.urgency	Medium is most common.
Fluency split	fluent 380; non-fluent 120	ticket metadata	Historical non-fluent FCR/CSAT do not support Sofia's concern.
Mean historical	422 minutes; median 214	history.resolution_minut	This is resolution time,

resolution time		es	not time to first reply.
Live response time	Not available	No first-reply event field	Cannot establish the SLA baseline or improvement.
Top exact body frequency	14 tickets mention rollback revision question	Text frequency analysis	Repeated wording suggests a useful regression case.
Current 80-ticket validation outcome	8 auto-respond, 65 escalate, 7 blocked	single run 2026-11-02; metrics.json	Only 10% auto response; retain safety-first threshold.

Two rows matter more than the rest

The proportion of tickets already answerable from existing documentation tells you whether CloudServe lack answers or merely lack a way of reaching them. Those are different problems with different solutions. The non-fluent segment matters for a different reason: something in the transcripts points at it, and the data will tell you whether that person was right.

3 Section three: where the time actually goes

Understanding volume is not the same as understanding effort. A category that accounts for five per cent of tickets but forty per cent of agent hours is where the real problem lives. Work out, as far as your evidence allows, how effort is distributed.

Ticket category	Share of volume	Estimated share of effort	Why the two differ	Evidence for your estimate
Routine docs questions	Part of 71.4% answerable share	45% planning estimate	Repetitive work may be fast individually	Qualitative only; no measured time by ticket category.
Security/ compliance/data residency	Counts available in labels; no task time	25% planning estimate	High risk and review/escalation increase effort	Risk/review likely add effort; no time study.
Novel/unclear incidents	unclear and unsupported cases present	20% planning estimate	Diagnosis and handoff likely take longer	Interview themes only; no time study.
All other ticket work	Remaining mix	10% planning estimate	Cross-channel context varies	Residual estimate; no agent time data.

The steps an agent takes on a typical ticket

Sofia describes her working process in some detail in the second transcript. Reconstruct it as a sequence of steps and record roughly where the time goes at each one. The steps that sound trivial are frequently where the minutes accumulate.

Step	What the agent does	Roughly how long	Could this be automated? Say why or why not
1	Read and normalize ticket	2–4 min (est.)	Partly: parse metadata; preserve original text.
2	Classify intent and	1–2 min (est.)	Assistive only; urgency

	urgency		accuracy must improve.
3	Search approved documentation	3–8 min (est.)	Yes, retrieval is automated; validate relevance.
4	Check risk, confidence and support	2–5 min (est.)	Automate gates; send uncertain/high risk to human.
5	Draft sourced reply or prepare handoff	3–8 min (est.)	Drafting optional provider; do not invent missing facts.
6	Agent reviews, replies and logs outcome	2–5 min (est.)	Human remains responsible; capture actual reply time and outcome.
7	Resolve/follow up and update docs	5–15 min (est.)	No autonomous resolution/knowledge edits in scope.

4 Section four: what the client counts as success

You cannot improve a number nobody is watching. Marcus names the figures he reports and is explicit about which one he actually cares about, and they are not the same. Record what CloudServe measure, who watches each measure, and what would have to change for this project to be considered worthwhile.

Measure	Who watches it	Current value	What they would call success	How confident are you in this?
First-contact resolution	Marcus / support lead	43.8% history	>=60% target in main instructions	Medium: labeled synthetic baseline.
Time to first reply	Support lead	Not in dataset; brief narrative 8–12h	<5 minutes automation target; 2h contractual SLA	Low: baseline not instrumented.
CSAT	Support lead	2.972/5 historical	>=4.0 target	Medium-low: synthetic historical rating.
Escalation rate	Support lead	189/500 expected route =37.8%	<=30% target	Medium: expected route label, not system outcome.
Intent confidence calibration	ML/system owner	ECE 16.64% on validation	<=5 percentage points	Low: one 80-ticket sample.
Citation / grounding safety	Governance owner	Automated proxies; human audit pending	>=95% citation accuracy; hallucination <=5%	Low: human review not performed.
p95 processing latency	Engineering owner	6.49ms offline local	<3 seconds	Medium: offline local only; no provider call.

The sentence test

Complete the following sentence using only figures you recorded above. If you cannot complete it, your discovery is not finished.

Prompt	Your answer
This project will have been worth doing if, within three months of launch, …	A monitored, human-reviewed pilot demonstrates safer source-based replies and improves FCR, CSAT and SLA performance against measured live

	baselines.
We will know it did not work if we see ...	Unsupported or harmful replies, poor urgency/calibration, persistent quality gaps, no measurable business lift, or unsafe human handoff.
The measure the client will actually be judged on internally is ...	FCR primary; response time and CSAT as safeguards.

5 Section five: data, constraints and risk

Everything you build depends on data that already exists, and every piece of that data carries constraints. Record what exists, what condition it is in, and what you are not permitted to do with it. Ines and Daniel both raise concerns about data quality that belong in this table.

Data source	What it contains	Quality and gaps	Access constraints	Private data present?
Support tickets	500 dev + 80 visible validation tickets with labels/history	Synthetic; no live events or true first-response time	Local pack; validation should be used once as directed	No private customer identities expected; inspect before production.
Documentation	29 articles with IDs and text	Coverage labels; no independent content-quality audit	Approved corpus for prototype	No direct personal data indicated.
Stakeholder interviews	Five fictional transcript records	Qualitative, small sample; conflicting estimates	Capstone pack	No real interviewees.
Ground truth responses	200 reference answers, expected citations/claims	Not all may map to validation tickets; no dual human ratings performed	Use for evaluation only	Review text for any sensitive content before external use.
System decision log	Per-ticket action/prediction/source/guardrail details	Local SQLite; synthetic run records	Local file access control needed	Could hold ticket content; protect and define retention.
Environment/ provider config	Provider selection and secret keys	Keys must stay outside repo/logs	Local .env ignored	Secrets are private; never package.
Personal effort log	Dates, task notes, actual hours	Not yet supplied by student	Student completes from actual records	Personal data.

Initial risk register

List what could go wrong once this system is speaking to customers without supervision. You will develop these further in stage five, but the first pass belongs here, while the problem is still fresh and you have not yet become attached to a particular design.

What could go	How likely	How bad	Who it affects	First thought on

wrong				preventing it
The system answers confidently and incorrectly	Medium	High	Customers, support, company	Confidence and grounding gates, approved citations, human audit, kill switch.
Private information appears in a reply	Low/unknown	Critical	Customers/ company	PII detection, minimal context, block-and-escalate, secure logs.
Prompt injection in ticket text	Medium	High	Customers/ company	Treat ticket as untrusted data; delimit, instruction integrity checks; test adversarially.
Urgency misclassified	High in validation (56.3% wrong)	High	Customers especially urgent cases	Do not automate priority; human triage; improve labeled data and evaluate.
Unequal quality by group	Unknown	High	Non-fluent, regional, tier groups	Human quality audits by group; minimum sample; monitor and pause if gap.
Model provider unavailable	Medium	Medium	Agents/customers	Deterministic fallback to escalation; no answer on unavailable generation.
Secrets/logs exposed	Low	High	Customers/ company	Env secrets, least-privilege access, redaction and retention policy.
Bad release or drift	Medium	High	All users	Versioned prompts/models, monitored rollout,

				kill switch and rollback.

6 Section six: the problem statement

Now, and only now, write down what the problem actually is. Every clause of what you write must be supported by a row you have already filled in, and you are asked to cite those rows explicitly. This is the paragraph your entire project rests on.

Element	Your statement	Which section and row supports it
What the client asked for	A chatbot that can respond to support tickets.	Brief; interview transcripts.
What the evidence suggests they actually need	A safe support-assistance workflow that classifies, retrieves approved evidence, answers only when supported, and prepares clear human handoffs; with measured outcomes and governance.	Tables 2–5 and 9–12; validation evaluation.
What the first build can responsibly do	Prototype with low auto-response coverage, transparent escalation, local decision logging, optional provider drafting and kill switch. Not production-ready.	Validation metrics; source code; risk register.
What remains unproven	Live SLA/CSAT/FCR lift, human-rated answer quality, broad fairness, production resilience.	Missing telemetry and human audit.
What will change if this is solved	Faster sourced replies; safer handoffs.	Track live FCR, reply time and CSAT.
What is explicitly not in scope	No autonomous account changes, production launch or agent replacement.	Safety/scope boundary.

The one paragraph version

Write the problem in a single paragraph, in language the head of support would recognise and agree with. No technical vocabulary. If you cannot express it without mentioning retrieval or embeddings, you are describing your solution rather than their problem.

Problem statement
CloudServe's support function faces slow replies, low first-contact resolution and poor customer satisfaction. Development data suggests many tickets map to existing documentation (71.4%), but this does not prove customers or agents can find the answers. The project therefore builds a human-supervised support workflow that classifies each ticket, retrieves approved evidence, drafts a cited answer only when confidence and grounding gates pass, and otherwise provides an uncertainty-aware escalation. The prototype must be judged

on safety, measured business outcomes and subgroup quality; current offline results show it is not ready for unattended production.

Before you move to stage two

Read your problem statement back against each of the five transcripts in turn and ask whether that person would recognise it as a description of their situation. If any of the five would object, either your framing is off or you have found something they cannot see, and you should be able to say which. A statement that all five would simply agree with is usually one that has not gone far enough.